

Data Flow Diagram Specification

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Team

Data Flow Diagram (DFD) and Legend

This specification details how financial applicant data moves through the system processes during credit scoring.

Symbol	Name	Smart Lender Implementation
Oval	External Entity	Applicant (enters credit info), Financial Analyst (uploads batch CSVs)
Rectangle (Header)	Process	1. Impute & preprocess features, 2. Apply scaler, 3. Score XGBoost, 4. Evaluate risk flags
Rectangle (Solid)	Data Store	train.csv (source dataset), rdf.pkl (serialized pipeline assets), temp/predictions_output.csv
Arrow	Data Flow	Inputs data flow from forms to backend, predictions data flow to clients

DFD Process Flow Chart

1. Applicant Entity -> [Single Form / JSON] -> Process 1.0: Validate & Preprocess
2. Process 1.0 -> [Fills missing with Mode/Median, encodes, log transform] -> Process 2.0: Feature Scaler
3. Process 2.0 -> [StandardScaler transform] -> Process 3.0: Model Score Engine
4. Data Store: rdf.pkl -> [Loads XGBoost & Scaler] -> Process 3.0
5. Process 3.0 -> [Predict probability] -> Process 4.0: Risk Flags Evaluator
6. Process 4.0 -> [Trigger poor credit / low income / DTI flags] -> External Entity: Web Client